

Military aviation: Daniel K. Inouye Fighter Squadron & Maintenance Facility New Megadoor hangar door installation

Megadoor vertical lifting fabric doors contribute to highest honors in the USAF Design Awards

US Joint Base Pearl Harbor-Hickam, on the Hawaiian island of Oahu, was created in 2010 as part of the Defense Base Realignment and Closure (BRAC) Congressional Act. The base is a merger of Naval Station Pearl Harbor with Hickam Air Force Base, which has been a national key to Pacific mobility since 1948.

Among the base's major tenants is the 154th Wing under the Hawaii Air National Guard (ANG), which flies stealthy and maneuverable F-22 Raptors. In 2013, aligned with the base's vision of the highest-quality facilities, the ANG commissioned what would be an award-winning new hangar for these emblematic fighter jets.

Customer challenges

Like all US military building projects, the new hangar design required compliance with the Whole Building Design Guide, which serves many industries but was originally designed for US Department of Defense. In addition, the military has its own United Facilities Guide Specifications (UFGS), which are used in specifying construction for the military services. Energy efficiency is a key target in both, in part due to the Energy Independence and Security Act, a government mandate requiring many new and renovated federal buildings to meet "Net-Zero" energy goals by 2020.

For Burns & McDonnell, the designer of record for the project, the hangar doors were a key aspect in meeting these guidelines. The decision involved not only reducing energy use, but also making maximum use of the real estate. Since the hangar would be used for maintenance, there were also more practical considerations, such as avoiding the inconvenience of moving 500 lb tool carts over door rails, which are required with traditional bottom rolling door technology.

What we provided

As a global leader in working with architects and military end users, ASSA ABLOY Entrance Systems supported Burns & McDonnell in defining and evaluating the UFGS 08 34 16.20 Vertical Lift Fabric Door specification. For the final design, Burns & McDonnell chose vertical lifting fabric doors over traditional bottom rolling doors.

The two 174' x 25' Megadoor vertical lifting fabric doors used on the project combine fast operating speeds with minimized space requirements. They eliminate not only the need for door pockets, but also the need for a costly grade beam and rails – all of which would be required with traditional bottom rolling hangar doors.

The Megadoor vertical lifting fabric doors also provide exceptional energy efficiency. The doors' translucent fabric allows ambient natural light into the hangar, reducing the need for electric light and creating a more productive working environment within. Furthermore, the doors can be partially opened when moving equipment into or out of the hangar, which minimizes the loss of conditioned air. This is not possible with traditional bottom rolling doors, which expose the full height of the opening and quickly result in the loss of all conditioned air.

Benefits

Since its completion, the Hawaii ANG's new F-22 hangar has received praise both on base and off. In fact, the new facility earned a top-ranking Honor Award in the 2014 USAF Design Awards. These awards were established in 1976 as a recognition and benchmark of design excellence.

Central to the jury's motivation were the hangar's sustainable features and low maintenance. The hangar doors play a key role in both aspects, thanks to energy-efficient sealing and an almost maintenance-free design with few moving parts. These characteristics are represented in more than 600 Megadoor installations for the US military – and will be even more in the spotlight following the recent award.



Testimonial
"The success of the Daniel K. Inouye Fighter Squadron & Aircraft Maintenance Facility was definitively elevated as the direct result of a fully collaborative team of professionals, including our client, the Hawaii Air National Guard, the designers Burns & McDonnell, as well as all of our subcontractors and suppliers, including ASSA ABLOY Entrance Systems. FED-CON-1 A JV thanks them all for their efforts in delivering this premier facility." – Andrew Snow, Construction Manager FED-CON-1 A JV



Honors
"This is a very innovative facility and layout, with a lot of sustainable, energy-conscious features. Passive and active energy efficiency. Very well done." – Jury member, 2014 USAF Design Awards

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